

Definitions (straight-edged polygons)

A 1-1 transcription of Cam's handwritten **definitions** (no-rounding / sharp polygons). Everything in black is his; everything in blue is my own note. Vertices are lowercase $\vec{v}_k$ (matching the handwriting); $z(k)/z'(k)$ are the next/previous vertices CCW and $\varepsilon_{\alpha\beta}$ is the 2D Levi-Civita symbol with $\varepsilon_{12} = +1$.

1 Definitions

Our polygons have vertices $\vec{v}_k$ for $k \in [0, N_v)$, where N_v is the total number of vertices across all polygons in the packing. Let there be N polygons so that polygon K has n_K vertices; then $N_v = \sum_{K=1}^N n_K$. Let

$$\vec{e}_k \equiv \vec{v}_{z(k)} - \vec{v}_k, \quad (1.1)$$

$$\vec{b}_k \equiv \vec{v}_{z(k)} - \vec{v}_{z'(k)}, \quad (1.2)$$

and also (area of a polygon)

$$A = \frac{1}{2} \sum_{k=0}^{n-1} e_{z(k)}^\alpha \varepsilon_{\alpha\beta} e_k^\beta. \quad (1.3)$$

This is written with *edge* vectors $\vec{e}$, but $\frac{1}{2} \sum_k \vec{e}_{z(k)} \times \vec{e}_k$ is not the polygon area – it overcounts (e.g. it gives 2 for the unit square, whose area is 1). The area is the *vertex* shoelace $A = \frac{1}{2} \sum_k v_k^\alpha \varepsilon_{\alpha\beta} v_{z(k)}^\beta$ (CCW-positive), which is what `updateOverlapArea` uses. The perimeter is

$$P = \sum_{k=0}^{n-1} |\vec{e}_k|. \quad (1.4)$$

Let us also be very clear that the vertices are ordered CCW.

2 Intersections

Obviously $\vec{v}_k + \vec{e}_k = \vec{v}_{z(k)}$ by definition, so

$$\vec{P}_k(s) \equiv \vec{v}_k + s \vec{e}_k \quad \text{for } s \in [0, 1] \quad (2.1)$$

interpolates linearly between the two points. For two edges i and j which intersect, there exist $s, t \in [0, 1]$ with $\vec{P}_i(s) = \vec{P}_j(t)$. Rearranging gives $v_i^\alpha - v_j^\alpha + s e_i^\alpha - t e_j^\alpha = 0$, and eliminating t ,

$$\frac{v_i^x - v_j^x + s e_i^x}{e_j^x} - \frac{v_i^y - v_j^y + s e_i^y}{e_j^y} = 0,$$

$$s \left(\frac{e_i^x}{e_j^x} - \frac{e_i^y}{e_j^y} \right) = -\frac{v_i^x - v_j^x}{e_j^x} + \frac{v_i^y - v_j^y}{e_j^y},$$

$$s(e_i^x e_j^y - e_j^x e_i^y) = -[(v_i^x - v_j^x)e_j^y - (v_i^y - v_j^y)e_j^x],$$

i.e. $s e_i^\alpha \varepsilon_{\alpha\beta} e_j^\beta = (\vec{v}_j - \vec{v}_i)^\alpha \varepsilon_{\alpha\beta} e_j^\beta$, so

$$s = \frac{(\vec{v}_j - \vec{v}_i)^\alpha \varepsilon_{\alpha\beta} e_j^\beta}{e_i^\alpha \varepsilon_{\alpha\beta} e_j^\beta}, \quad (2.2)$$

and by symmetry

$$t = \frac{(\vec{v}_i - \vec{v}_j)^\alpha \varepsilon_{\alpha\beta} e_i^\beta}{e_j^\alpha \varepsilon_{\alpha\beta} e_i^\beta}. \quad (2.3)$$

These now match the handwriting (2.2)/(2.3) and the sign `polygonPairIntersections` uses ($s = s_A$, $t = s_B$); an earlier transcription here dropped the leading minus on the “ $s(\dots) = \dots$ ” line. A genuine intersection needs $s, t \in (0, 1)$.

3 Area of overlap

Let there be two intersecting polygons A and B . The area of intersection between A and B is defined as

$$|A \cap B| \equiv \frac{1}{2} \varepsilon_{\alpha\beta} \oint_{\partial(A \cap B)} X^\alpha dX^\beta \quad (3.1)$$

for the boundary of the polygon formed by the overlaps, $\partial(A \cap B)$. As you traverse the shape formed by this overlap, the vertices come in 4 flavors:

1. $\vec{v}_k \notin \partial(A \cap B)$, $\vec{v}_{z'(k)} \notin \partial(A \cap B)$
2. $\vec{v}_k \notin \partial(A \cap B)$, $\vec{v}_{z'(k)} \in \partial(A \cap B)$
3. $\vec{v}_k \in \partial(A \cap B)$, $\vec{v}_{z'(k)} \notin \partial(A \cap B)$
4. $\vec{v}_k \in \partial(A \cap B)$, $\vec{v}_{z'(k)} \in \partial(A \cap B)$

We may have the case where the number of vertices on the boundary $\partial(A \cap B)$ is very large. There is a way to combat this, but first we consider an intersection between edges i and j , denoted $\{i, j\}$, where there exist s, t such that the point $\vec{v}_i + s\vec{e}_i \in \partial(A \cap B)$ but $\vec{v}_i + t\vec{e}_i \notin \partial(A \cap B)$ for $s > t$, and similarly for j except $s < t$. If you follow the boundary starting at the intersection and following the boundary of the polygon associated with edge i , then eventually you will reach another intersection $\{k, l\}$ where k follows j and l follows i . Denote this intersection pair as $\{i, j; k, l\}$.

For an edge from $\vec{P}$ to $\vec{Q}$ we define the area contribution as

$$h(\vec{P}, \vec{Q}) \equiv \frac{1}{2} (P^\alpha - R^\alpha) \varepsilon_{\alpha\beta} (Q^\beta - R^\beta), \quad (3.2)$$

where $\vec{R}$ is a reference point. We usually use the lowest indexed vertex in the lower indexed polygon between A and B . Note that there are exactly as many pairs of intersections as intersections: every intersection has a follower.

Our *total* overlap area is then

$$U = \sum_{\{i,j;k,l\}} \left\{ \delta_{il} h(\vec{f}_{ij}, \vec{f}_{kl}) + (1 - \delta_{il}) \left[h(\vec{f}_{ij}, \vec{v}_{z(i)}) + h(\vec{v}_l, \vec{f}_{kl}) + \sum_{m=z(i)}^{z'(l)} h(\vec{v}_m, \vec{v}_{z(m)}) \right] \right\}, \quad (3.3)$$

$\vec{f}_{ij}$ is the intersection point of edges i, j . Same boundary walk as `polygonPairOverlapArea`, and the same shape as the rounded master's eq (4.6).

We can clean this up a little by defining a coordinate with a fractional or continuous index. Let i have a fractional part corresponding to the portion of the edge which has been traversed. Then $\vec{q}_i = \vec{v}_{i-} + (i - i^-)\vec{e}_{i-}$ (with i^- the integer part) and

$$U = \sum_{\{i,j;k,l\}} \left\{ \delta_{il} h(\vec{q}_i, \vec{q}_l) + (1 - \delta_{il}) \left[h(\vec{q}_i, \vec{v}_{z(i-)}) + h(\vec{v}_{l-}, \vec{q}_l) + \sum_{m=z(i-)}^{z'(l-)} h(\vec{v}_m, \vec{v}_{z(m)}) \right] \right\}. \quad (3.4)$$

Let the intersections be $\{s_{i-}, s_{j-}, j, i\}$ in sorted form from left to right for shapes s_{i-} and s_{j-} . We then search for $\{s_{j-}, s_{i-}, l, k\}$ via binary search to find the corresponding “next” intersection. We can just save that intersection's index instead of keeping another huge list. We can then define $\delta_{mp} = 1$ when an edge m is fully between intersections $\{i, j\}$ and $\{k, l\}$ for $p = \{i, j; k, l\}$, and $\delta_{mp} = 0$ otherwise; all p with $\delta_{mp} = 1$ can be found by binary (or linear) search. We then have

$$\begin{aligned} U_{\text{ex}} &= \sum_{\{i,j;k,l\}} \left\{ \delta_{il} h(\vec{q}_i, \vec{q}_l) + (1 - \delta_{il}) [h(\vec{q}_i, \vec{v}_{z(i-)}) + h(\vec{v}_{l-}, \vec{q}_l)] \right\}, \\ U_{\text{int}} &= \sum_{m=0}^{N_v-1} \sum_{p=\{i,j;k,l\}} \delta_{pm} h(\vec{v}_m, \vec{v}_{z(m)}), \\ U &= U_{\text{int}} + U_{\text{ex}}. \end{aligned} \quad (3.5)$$

The external/internal split for parallelization – end-cap pieces (U_{ex}) vs whole interior edges (U_{int}) – matching the rounded master's eq (4.7)/(4.8).

4 Forces on the boundary

Differentiating the integral over the boundary has a trick associated with it related to the Hadamard structure theorem: we can get out of computing the derivative of the intersection point (thank God). Take $A_\cap = \frac{1}{2} \oint_{\partial\Omega} X^\alpha \varepsilon_{\alpha\beta} dX^\beta$ for the boundary Ω and consider perturbing $\vec{X} \mapsto \vec{X} + \delta\vec{X}$, so that

$$\delta A_\cap = \frac{1}{2} \oint_{\partial\Omega} \delta X^\alpha \varepsilon_{\alpha\beta} dX^\beta + \frac{1}{2} \oint_{\partial\Omega} X^\alpha \varepsilon_{\alpha\beta} d\delta X^\beta.$$

Using integration by parts on the second term (the boundary term $\frac{1}{2} X^\alpha \varepsilon_{\alpha\beta} \delta X^\beta|_{\Omega^-}^{\Omega^+}$ vanishes on the closed loop) and $\varepsilon_{\alpha\beta} = -\varepsilon_{\beta\alpha}$,

$$\delta A_\cap = \oint_{\partial\Omega} \delta X^\alpha \varepsilon_{\alpha\beta} dX^\beta. \quad (4.1)$$

We then let $\vec{X}_m(\lambda) = \vec{v}_m + \lambda \vec{e}_m$ for a point on edge m , so $\delta \vec{X}_m(\lambda) = (1 - \lambda) \delta \vec{v}_m + \lambda \delta \vec{v}_{z(m)}$ and $d\vec{X}_m = \vec{e}_m d\lambda$. Then for the segments,

$$\begin{aligned} \delta A_\cap &= \sum_m \sum_\alpha \int_{\lambda_0^m}^{\lambda_f^m} [(1 - \lambda) \delta v_m^\alpha + \lambda \delta v_{z(m)}^\alpha] \varepsilon_{\alpha\beta} e_m^\beta d\lambda \\ &= \sum_m \sum_\alpha \left\{ [(\lambda_f - \lambda_0) - \frac{1}{2}(\lambda_f^2 - \lambda_0^2)] \delta v_m^\alpha + \frac{1}{2}(\lambda_f^2 - \lambda_0^2) \delta v_{z(m)}^\alpha \right\} \varepsilon_{\alpha\beta} e_m^\beta. \end{aligned}$$

Let $\bar{s}_m = \frac{1}{2}(\lambda_f + \lambda_0)$ and $\Delta s_m = \lambda_f - \lambda_0$ (so $\frac{1}{2}(\lambda_f^2 - \lambda_0^2) = \bar{s}_m \Delta s_m$), giving

$$\delta A_\cap = \sum_\alpha \sum_m [(1 - \bar{s}) \delta v_m^\alpha + \bar{s} \delta v_{z(m)}^\alpha] \varepsilon_{\alpha\beta} \Delta s_m e_m^\beta. \quad (4.2)$$

Writing this with Kronecker deltas, $\delta A_\cap = \sum_k \sum_\alpha [\sum_m ((1 - \bar{s}_m) \delta_{mk} + \bar{s}_m \delta_{z(m)k}) \Delta s_m \varepsilon_{\alpha\beta} e_m^\beta] \delta v_k^\alpha$, and since $\delta A_\cap = \sum_k \sum_\alpha (\partial A_\cap / \partial v_k^\alpha) \delta v_k^\alpha$, we conclude

$$\frac{\partial A_\cap}{\partial v_k^\alpha} = \sum_m [(1 - \bar{s}_m) \delta_{mk} + \bar{s}_m \delta_{z(m)k}] \Delta s_m \varepsilon_{\alpha\beta} e_m^\beta. \quad (4.3)$$

This is the overlap force for the sharp model (function 4 to come). It is the rounded master's eq (7.6) with the kiss-point Jacobians $\partial a^\pm / \partial v$ collapsed to plain vertices: an edge piece spanning $[\lambda_0, \lambda_f]$ deposits $(1 - \bar{s}) \Delta s$ of its cross term on $\vec{v}_m$ and $\bar{s} \Delta s$ on $\vec{v}_{z(m)}$. A full interior edge has $\bar{s} = \frac{1}{2}$, $\Delta s = 1$.

5 Intersections (pairing and ordering)

A second section titled “Intersections” in the handwriting – the ordering/pairing algorithm that supplies the $\{i, j; k, l\}$ pairs and the δ_{pm} above.

Consider an intersection $\{i, j\}$ which intersect at point $\vec{v}_i + s_i \vec{e}_i = \vec{v}_j + s_j \vec{e}_j$. Let us also consider the polygon index $\rho_i, \rho_j \in [0, N)$. WLOG, $\vec{v}_i + (s_i + \varepsilon) \vec{e}_i$ is on the overlap boundary for some $\varepsilon > 0$. Sort all of the intersections by $\{\rho_i, \rho_j, j, s_j, i, s_i\}$.

Then we find the intersection pair. For intersection $I_\alpha = \{\rho_i, \rho_j, j, s_j, i, s_i\}$ we want to find intersection $I_\beta = \{\rho_k, \rho_l, l, s_l, k, s_k\}$. We want $\rho_k = \rho_j$ and $\rho_i = \rho_l$, which should be a short list, and we can find its boundaries with binary search (in general it has 2 entries). We then choose β so that $|i + s_i - (l + s_l)|$ is minimal over all polygons.

Next we loop over *all* vertices and check all intersections of that shape to see if vertex $\vec{v}$ is between $i + s_i$ and $l + s_l$, where $v \neq l$. If it is, you can compute the components necessary for its gradient and energy.

6 Mollification

Unfortunately, this alone doesn't work. FIRE chokes and may or may not reach the basin. It certainly doesn't reach the minimum. We can't use a higher-order method because the forces aren't continuous, meaning the Hessian is ill-defined. We can resolve both issues by mollifying the boundaries.

Consider the area of overlap,

$$A_\cap = \int_{\mathbb{R}^2} \mathbf{1}_A \mathbf{1}_B d^2x. \quad (6.1)$$

We can consider a kernel K_σ which smooths out the edges, so that instead of a step boundary we have a smooth boundary. Applying this to *one* of the polygons gives

$$A_\cap^\sigma = \int_{\mathbb{R}^2} \mathbf{1}_A (K_\sigma * \mathbf{1}_B) d^2x \quad (6.2)$$

(we convolved with just one indicator), i.e.

$$A_\cap^\sigma = \int d^2x \int d^2x' \mathbf{1}_A(\vec{x}) K_\sigma(\vec{x} - \vec{x}') \mathbf{1}_B(\vec{x}'). \quad (6.3)$$

We need to be careful what K_σ we choose. If we choose a Gaussian, it'll be a nightmare. We know this is not realistic anyway, so we can just use the Plummer model.

The 2D surface density (from Wikipedia) is

$$K_\sigma = \frac{\sigma^2}{\pi(\sigma^2 + |\vec{r}|^2)^2}, \quad (6.4)$$

but of course it doesn't give the potential or anything. The vector field (ordinarily $\vec{F}$) is given by

$$\nabla \cdot \vec{E}_\sigma = K_\sigma \quad (6.5)$$

and

$$\nabla \Phi_\sigma = \vec{E}_\sigma. \quad (6.6)$$

Let

$$\Psi_B^\sigma(\vec{x}) \equiv (K_\sigma * \mathbf{1}_B)(\vec{x}), \quad (6.7)$$

or

$$\Psi_B^\sigma(\vec{x}) = \int d^2y K_\sigma(\vec{y} - \vec{x}) \mathbf{1}_B(\vec{y}) = \int d^2y (\nabla \cdot \vec{E}_\sigma)(\vec{y} - \vec{x}) \mathbf{1}_B(\vec{y}).$$

So from Gauss's theorem,

$$\Psi_B^\sigma(\vec{x}) = \oint_{\partial B} \vec{E}_\sigma(\vec{y} - \vec{x}) \cdot \hat{n}_B(\vec{y}) d\ell_y. \quad (6.8)$$

Since $\vec{E}_\sigma = \nabla \Phi_\sigma$, we have

$$\Psi_B^\sigma(\vec{x}) = \oint_{\partial B} (\nabla_{\vec{x}} \Phi_\sigma)(\vec{y} - \vec{x}) \cdot \hat{n}_B(\vec{y}) d\ell_y. \quad (6.9)$$

And again, from Wikipedia, the gradient theorem is $\int_\Omega \nabla u dV = \int_{\partial\Omega} u \vec{\nu} dS$, where $\vec{\nu}$ is the outward-pointing unit normal to $\partial\Omega$. Now

$$A_\cap^\sigma = \int d^2x \Psi_B^\sigma(\vec{x}) = \int d^2x \oint_{\partial B} (\nabla_{\vec{x}} \Phi_\sigma)(\vec{y} - \vec{x}) \cdot \hat{n}_B(\vec{y}) d\ell_y,$$

and using the gradient theorem,

$$A_\cap^\sigma = \oint_{\partial A} d\ell_x \oint_{\partial B} d\ell_y [-\Phi_\sigma(|\vec{y} - \vec{x}|)] \hat{n}_A(\vec{x}) \cdot \hat{n}_B(\vec{y}). \quad (6.10)$$

Sections §6–§7 re-derive, from scratch, the Plummer mollification the code uses; (6.10) is exactly the double-boundary reduction in `plummerOverlap.tex` (its eq. (6)). I silently tidied a few integration-variable slips (e.g. $\mathbf{1}_B(\vec{x}) \rightarrow \mathbf{1}_B(\vec{y})$, and the second measure $d^2y \rightarrow d^2x'$).

7 Getting $\Phi_\sigma(r)$

We noted $\nabla \cdot \vec{E} = K_\sigma$, which means

$$\vec{E}_\sigma(\vec{r}) = C \frac{\vec{r}}{\sigma^2 + |\vec{r}|^2}. \quad (7.1)$$

Quick check:

$$\frac{\partial}{\partial x} \left(C \frac{x}{\sigma^2 + |\vec{r}|^2} \right) = C \frac{\sigma^2 + y^2 - x^2}{(\sigma^2 + |\vec{r}|^2)^2},$$

so by symmetry

$$\nabla \cdot \vec{E}_\sigma = C \frac{(\sigma^2 + y^2 - x^2) + (\sigma^2 + x^2 - y^2)}{(\sigma^2 + |\vec{r}|^2)^2} = 2\pi C \frac{\sigma^2}{\pi(\sigma^2 + |\vec{r}|^2)^2} = 2\pi C K_\sigma,$$

and $\nabla \cdot \vec{E}_\sigma = K_\sigma$ forces $C = \frac{1}{2\pi}$. Therefore

$$\vec{E}_\sigma(\vec{r}) = \frac{\vec{r}}{2\pi(|\vec{r}|^2 + \sigma^2)}. \quad (7.2)$$

Next, $\nabla \Phi_\sigma = \vec{E}_\sigma(\vec{r})$, so trying $\Phi_\sigma = C \ln(r^2 + \sigma^2)$ gives $\nabla \Phi_\sigma = (C \frac{2x}{r^2 + \sigma^2}, C \frac{2y}{r^2 + \sigma^2}) = 2C \frac{\vec{r}}{r^2 + \sigma^2}$; matching (7.2) needs $2C = \frac{1}{2\pi}$, i.e. $C = \frac{1}{4\pi}$, and

$$\Phi_\sigma(r) = \frac{1}{4\pi} \ln(r^2 + \sigma^2). \quad (7.3)$$

Combining (6.10) and (7.3) gives

$$A_\cap^\sigma = -\frac{1}{4\pi} \oint_{\partial A} d\ell_x \oint_{\partial B} d\ell_y \ln(|\vec{x} - \vec{y}|^2 + \sigma^2) \hat{n}_A \cdot \hat{n}_B. \quad (7.4)$$

The handwriting still cites this as “combining 6.10 and 6.13”; with §7 now renumbered, the potential (7.3) is 7.3, so I reference it by label. We can apply this to our polygons to get

$$A_\cap^\sigma = -\frac{1}{4\pi} \sum_{\alpha \in \partial A} \sum_{\beta \in \partial B} (\hat{n}_\alpha \cdot \hat{n}_\beta) \int_{e_\alpha} d\ell_x \int_{e_\beta} d\ell_y \ln(|\vec{x} - \vec{y}|^2 + \sigma^2). \quad (7.5)$$

Now what is $\int_{e_\alpha} d\ell_x \ln(|\vec{x} - \vec{y}|^2 + \sigma^2)$? With $d\ell_x = |\vec{e}_\alpha| dt$ and $\vec{x} = \vec{v}_\alpha + t \vec{e}_\alpha$ this is

$$|\vec{e}_\alpha| \int_0^1 dt \ln(|\vec{v}_\alpha - \vec{y}|^2 + \sigma^2 + 2(\vec{v}_\alpha - \vec{y}) \cdot \vec{e}_\alpha t + |\vec{e}_\alpha|^2 t^2).$$

Let

$$a = |\vec{e}_\alpha|^2, \quad b = 2(\vec{v}_\alpha - \vec{y}) \cdot \vec{e}_\alpha, \quad c = |\vec{v}_\alpha - \vec{y}|^2 + \sigma^2, \quad (7.6)$$

so $I = \int_0^1 dt \ln(at^2 + bt + c)$. With $u = \ln(at^2 + bt + c)$ and $dv = dt$,

$$I = t \ln(at^2 + bt + c) \Big|_0^1 - \int_0^1 dt \frac{2at^2 + bt}{at^2 + bt + c} = \ln(a + b + c) - \int_0^1 dt \left[2 - \frac{bt + 2c}{at^2 + bt + c} \right].$$

We want to isolate the $2at + b$ term.

Now we get rid of the bt by splitting $bt + 2c = \frac{b}{2a}(2at + b) + (2c - \frac{b^2}{2a})$ against $q' = 2at + b$, so that $\int_0^1 \frac{2at+b}{q} dt = \ln \frac{a+b+c}{c}$, giving

$$I = \ln(a + b + c) - 2 + \frac{b}{2a} \ln(a + b + c) - \frac{b}{2a} \ln c + \left(2c - \frac{b^2}{2a} \right) \int_0^1 \frac{dt}{at^2 + bt + c}.$$

Let

$$D \equiv \sqrt{4ac - b^2}. \quad (7.7)$$

We also let

$$\vec{w} \equiv \vec{v}_\alpha - \vec{y}, \quad (7.8)$$

so that $4ac - b^2 = 4|\vec{e}_\alpha|^2(|\vec{w}|^2 + \sigma^2) - 4(\vec{w} \cdot \vec{e}_\alpha)^2 = 4|\vec{e}_\alpha \times \vec{w}|^2 + 4|\vec{e}_\alpha|^2\sigma^2$, which means D is always real, and

$$I = \left(1 + \frac{b}{2a}\right) \ln(a + b + c) - 2 - \frac{b}{2a} \ln c + \frac{D}{a} \left[\arctan \frac{2a+b}{D} - \arctan \frac{b}{D} \right]. \quad (7.9)$$

Now matching the handwritten 7.9: the arctangent coefficient is $\frac{D}{a}$. The bracket is $\int_0^1 \frac{dt}{q} = \frac{2}{D} \left[\arctan \frac{2a+b}{D} - \arctan \frac{b}{D} \right]$, and its prefactor $2c - \frac{b^2}{2a} = \frac{D^2}{2a}$ combines to $\frac{D^2}{2a} \cdot \frac{2}{D} = \frac{D}{a}$ – the verified panel in `plummerOverlap.tex`.

The whole overlap is then

$$A_\cap^\sigma = -\frac{1}{4\pi} \sum_{\alpha \in \partial A} \sum_{\beta \in \partial B} (\hat{n}_\alpha \cdot \hat{n}_\beta) |\vec{e}_\alpha| \int_{e_\beta} d\ell_y I(\vec{y}),$$

and we integrate over $\vec{y}(s) = \vec{v}_\beta + \vec{e}_\beta s$, so that with $\vec{\Delta} \equiv \vec{v}_\alpha - \vec{v}_\beta$ and $\vec{w}(s) = \vec{\Delta} - s\vec{e}_\beta$,

$$\begin{aligned} b(s) &= 2\vec{\Delta} \cdot \vec{e}_\alpha - 2\vec{e}_\alpha \cdot \vec{e}_\beta s, \\ c(s) &= |\vec{\Delta}|^2 - 2s\vec{\Delta} \cdot \vec{e}_\beta + s^2|\vec{e}_\beta|^2 + \sigma^2. \end{aligned} \quad (7.10)$$

Then

$$D(s) = 2\sqrt{D_a s^2 + D_b s + D_c}, \quad (7.11)$$

and, with $I_2 = \int_{e_\beta} d\ell_y I(\vec{y})$,

$$\begin{aligned} \frac{I_2}{|\vec{e}_\beta|} &= \int_0^1 ds \left(1 + \frac{b(s)}{2a}\right) \ln(a + b(s) + c(s)) - 2 - \int_0^1 ds \frac{b(s)}{2a} \ln c(s) \\ &\quad + \int_0^1 ds \frac{D(s)}{a} \left[\arctan \frac{2a+b(s)}{D(s)} - \arctan \frac{b(s)}{D(s)} \right]. \end{aligned} \quad (7.12)$$

Arctangent coefficient corrected to $\frac{D(s)}{a}$ as above, and the outer Jacobian $L_\beta = |\vec{e}_\beta|$ (from $d\ell_y = |\vec{e}_\beta| ds$) is now carried by writing $I_2/|\vec{e}_\beta|$ on the left; the inner $|\vec{e}_\alpha|$ was restored earlier.

Carrying the outer Jacobian (recall $a = |\vec{e}_\alpha|^2$ is constant in s), the handwriting (its 7.13) splits this into

$$\frac{I_2}{|\vec{e}_\beta|} = I_a - 2 - I_b + I_c - I_d, \quad (7.13)$$

with

$$\begin{aligned} I_a &= \int_0^1 \left(1 + \frac{b(s)}{2a}\right) \ln(a + b(s) + c(s)) ds, & I_c &= \int_0^1 \frac{D(s)}{a} \arctan \frac{2a+b(s)}{D(s)} ds, \\ I_b &= \int_0^1 \frac{b(s)}{2a} \ln c(s) ds, & I_d &= \int_0^1 \frac{D(s)}{a} \arctan \frac{b(s)}{D(s)} ds. \end{aligned}$$

The four pieces need only the handwritten assumptions $|\vec{e}_\alpha| > 0$, $\sigma > 0$, and $e_\beta^x, e_\beta^y, \Delta^x, \Delta^y, s, t \in \mathbb{R}$ ($\sigma > 0$ alone keeps every logarithm argument and radicand positive, so no sign conditions are needed).

To evaluate them we rotate the frame so that $\vec{e}_\alpha = (|\vec{e}_\alpha|, 0)$. Then, with $\vec{\Delta} = \vec{v}_\alpha - \vec{v}_\beta$,

$$b(s) = 2|\vec{e}_\alpha|(\Delta^x - e_\beta^x s), \quad c(s) = |\vec{\Delta} - \vec{e}_\beta s|^2 + \sigma^2, \quad D(s) = 2|\vec{e}_\alpha| \sqrt{(\Delta^y - e_\beta^y s)^2 + \sigma^2}, \quad (7.14)$$

and the log argument completes the square in x ,

$$a + b(s) + c(s) = (|\vec{e}_\alpha| + \Delta^x - e_\beta^x s)^2 + (\Delta^y - e_\beta^y s)^2 + \sigma^2,$$

which is $c(s)$ with $\Delta^x \rightarrow |\vec{e}_\alpha| + \Delta^x$; likewise $1 + \frac{b}{2a} = (|\vec{e}_\alpha| + \Delta^x - e_\beta^x s)/|\vec{e}_\alpha|$ and $\frac{b}{2a} = (\Delta^x - e_\beta^x s)/|\vec{e}_\alpha|$ differ only by that shift. This exposes a common structure: with

$$J_1(p, s) \equiv \frac{p - e_\beta^x s}{|\vec{e}_\alpha|} \ln[(p - e_\beta^x s)^2 + (\Delta^y - e_\beta^y s)^2 + \sigma^2], \quad (7.15)$$

we have $\int_0^1 J_1(|\vec{e}_\alpha| + \Delta^x, s) ds = I_a$ and $\int_0^1 J_1(\Delta^x, s) ds = I_b$, and with

$$J_2(p, s) \equiv \frac{D(s)}{|\vec{e}_\alpha|^2} \arctan \frac{2|\vec{e}_\alpha|(p - e_\beta^x s)}{D(s)}, \quad (7.16)$$

we have $\int_0^1 J_2(|\vec{e}_\alpha| + \Delta^x, s) ds = I_c$ and $\int_0^1 J_2(\Delta^x, s) ds = I_d$.

Now change variables to normalize the softening: let

$$X \equiv \frac{1}{\sigma}(\Delta^y - e_\beta^y s), \quad s = \frac{1}{e_\beta^y}(\Delta^y - \sigma X), \quad ds = -\frac{\sigma}{e_\beta^y} dX, \quad (7.17)$$

and define the slope

$$m \equiv \frac{e_\beta^x}{e_\beta^y} \quad (7.18)$$

and scaled intercept

$$\nu \equiv \frac{1}{\sigma}(p - m \Delta^y), \quad (7.19)$$

so that $p - e_\beta^x s = \sigma(\nu + mX)$. The J_1 argument becomes $\sigma^2[(\nu + mX)^2 + X^2 + 1]$, giving

$$J_1(\nu, X) = \frac{2\sigma(\nu + mX)}{|\vec{e}_\alpha|} \ln \sigma + \frac{\sigma(\nu + mX)}{|\vec{e}_\alpha|} \ln F(X), \quad F(X) \equiv (1 + m^2)X^2 + 2m\nu X + (1 + \nu^2). \quad (7.20)$$

With $X_0 \equiv \Delta^y/\sigma$ and $X_1 \equiv \frac{1}{\sigma}(\Delta^y - e_\beta^y)$ (the images of $s = 0, 1$), the log integral is

$$J_{\log}(\nu) = -\frac{\sigma}{e_\beta^y} \int_{X_0}^{X_1} \left[\frac{2\sigma \ln \sigma}{|\vec{e}_\alpha|} + \frac{\sigma}{|\vec{e}_\alpha|} \ln F(X) \right] (\nu + mX) dX, \quad (7.21)$$

with $I_a = J_{\log}(\nu)$ at $\nu = \frac{1}{\sigma}(|\vec{e}_\alpha| + \Delta^x - m\Delta^y)$ and $I_b = J_{\log}(\nu)$ at $\nu = \frac{1}{\sigma}(\Delta^x - m\Delta^y)$. The handwritten 7.21 now carries the Jacobian $-\sigma/e_\beta^y$ on *both* terms (an earlier draft dropped the sign on the second). I verified (7.14)–(7.21) against the direct $\int_0^1 (1 + \frac{b}{2a}) \ln(a+b+c) ds$ and $\int_0^1 \frac{b}{2a} \ln c ds$ to $\sim 10^{-13}$. The payoff: with $A = 1 + m^2$, $B = 2m\nu$, $C = 1 + \nu^2$ the discriminant $4AC - B^2 = 4(1 + m^2 + \nu^2) > 0$ *always*, so the antiderivative is a real arctan – no branch conditions, unlike the direct attack on (7.13).

Doing the elementary $\ln \sigma$ integral, $\int_{X_0}^{X_1} (\nu + mX) dX = \nu(X_1 - X_0) + \frac{m}{2}(X_1^2 - X_0^2)$, the handwriting (7.22) writes

$$J_{\log}(\nu) = -\frac{\sigma^2}{|\vec{e}_\alpha| e_\beta^y} \left\{ \left[\nu(X_1 - X_0) + \frac{m}{2}(X_1^2 - X_0^2) \right] \ln \sigma^2 + \int_{X_0}^{X_1} (\nu + mX) \ln(AX^2 + BX + C) dX \right\}. \quad (7.22)$$

Sign of the handwritten 7.22: the overall factor must stay $-\sigma^2/(|\vec{e}_\alpha| e_\beta^y)$ (the latest draft flipped it to +, with both brace terms +). Only the minus matches the direct I_a, I_b (to $\sim 10^{-13}$; the plus misses by $O(10)$).

Closing I_a, I_b (my color). The remaining integral is elementary and real for $4AC - B^2 > 0$: with the primitives $\Lambda_0^F = \int \ln F dX$, $\Lambda_1^F = \int X \ln F dX$,

$$\begin{aligned}\Lambda_0^F(X) &= \left(X + \frac{B}{2A}\right) \ln F - 2X + \frac{\sqrt{4AC-B^2}}{A} \arctan \frac{2AX+B}{\sqrt{4AC-B^2}}, \\ \Lambda_1^F(X) &= \left(\frac{X^2}{2} - \frac{B^2-2AC}{4A^2}\right) \ln F - \frac{X^2}{2} + \frac{BX}{2A} - \frac{B\sqrt{4AC-B^2}}{2A^2} \arctan \frac{2AX+B}{\sqrt{4AC-B^2}},\end{aligned}$$

one has $\int_{X_0}^{X_1} (\nu + mX) \ln F dX = [\nu \Lambda_0^F + m \Lambda_1^F]_{X_0}^{X_1}$; substituted into (7.22) this gives I_a ($\nu = \frac{1}{\sigma}(|\vec{e}_\alpha| + \Delta^x - m\Delta^y)$) and I_b ($\nu = \frac{1}{\sigma}(\Delta^x - m\Delta^y)$), checked to $\sim 3 \times 10^{-13}$. (Mathematica returns the same in the ArcTanh branch; it is real because $\sqrt{B^2 - 4AC} = i\sqrt{4AC - B^2}$ and $\text{ArcTanh}(iz) = i \arctan z$, so the i 's cancel.)

For J_2 the substitution (7.17) gives

$$D(X) = 2|\vec{e}_\alpha|\sigma\sqrt{X^2+1}, \quad (7.23)$$

and $p - e_\beta^x s = \sigma(\nu + mX)$, so $J_2(p, X) = \frac{2\sigma}{|\vec{e}_\alpha|} \sqrt{X^2+1} \arctan \frac{\nu + mX}{\sqrt{X^2+1}}$ and

$$I_{\tan}(p) = -\frac{2\sigma^2}{|\vec{e}_\alpha|e_\beta^y} \int_{X_0}^{X_1} \sqrt{X^2+1} \arctan \frac{\nu + mX}{\sqrt{X^2+1}} dX, \quad (7.24)$$

with $I_c = I_{\tan}(|\vec{e}_\alpha| + \Delta^x)$ and $I_d = I_{\tan}(\Delta^x)$ (checked to $\sim 10^{-14}$). Integrating by parts with $u = \arctan \frac{\nu+mX}{\sqrt{X^2+1}}$ ($du = \frac{m-\nu X}{(AX^2+BX+C)\sqrt{X^2+1}} dX$) and $dV = \sqrt{X^2+1} dX$,

$$\begin{aligned}I_{\tan}(p) &= -\frac{2\sigma^2}{|\vec{e}_\alpha|e_\beta^y} \left\{ \arctan \frac{\nu+mX}{\sqrt{X^2+1}} \int \sqrt{X^2+1} dX \Big|_{X_0}^{X_1} \right. \\ &\quad \left. - \int_{X_0}^{X_1} \left(\int \sqrt{X^2+1} dX \right) \frac{m-\nu X}{\sqrt{X^2+1} (AX^2+BX+C)} dX \right\},\end{aligned} \quad (7.25)$$

with $\int \sqrt{X^2+1} dX = \frac{1}{2}(X\sqrt{X^2+1} + \text{arcsinh } X)$; since $X\sqrt{X^2+1}/\sqrt{X^2+1} = X$ collapses the first inner integral to a rational integrand, this becomes (handwriting 7.26)

$$\begin{aligned}I_{\tan}(p) &= -\frac{2\sigma^2}{|\vec{e}_\alpha|e_\beta^y} \left\{ \left[\frac{1}{2} \arctan \frac{\nu+mX}{\sqrt{X^2+1}} (X\sqrt{X^2+1} + \text{arcsinh } X) \right]_{X_0}^{X_1} \right. \\ &\quad + \frac{1}{2} \int_{X_0}^{X_1} \frac{\nu X^2 - mX}{AX^2 + BX + C} dX \\ &\quad \left. + \frac{1}{2} \int_{X_0}^{X_1} \frac{\text{arcsinh } X}{\sqrt{X^2+1}} \frac{\nu X - m}{AX^2 + BX + C} dX \right\}.\end{aligned} \quad (7.26)$$

Finishing $I_{\tan}$ (my color). The boundary term is explicit; the rational (middle) integral of (7.26) is elementary (polynomial division leaves a $\ln(AX^2+BX+C)$ plus an $\arctan \frac{2AX+B}{\sqrt{4AC-B^2}}$, real since $4AC - B^2 = 4(1 + m^2 + \nu^2)$). The last, arcsinh, integral $J_{\text{arcsinh}} \equiv \frac{1}{2} \int_{X_0}^{X_1} \frac{\text{arcsinh } X}{\sqrt{X^2+1}} \frac{\nu X - m}{AX^2 + BX + C} dX$ is the remaining dilog piece.

Evaluating J_{arcsinh} : the fully-real route (my derivation). No trick removes the dilogarithm $-4AC - B^2 = 4(1 + m^2 + \nu^2) > 0$, so $AX^2 + BX + C$ has a complex-conjugate root pair for all m, ν and the integral is genuinely dilog-level. The cleanest path keeps everything manifestly real. With $X = \sinh t$ ($t_0 = \text{arcsinh } X_0$, $t_1 = \text{arcsinh } X_1$, so $\text{arcsinh } X = t$ and $dX/\sqrt{X^2+1} = dt$),

$$J_{\text{arcsinh}} = \frac{1}{2} \int_{t_0}^{t_1} t \frac{\nu \sinh t - m}{A \sinh^2 t + B \sinh t + C} dt. \quad (7.27)$$

Integrate by parts with $u = t$, $dv = (\nu \sinh t - m)/Q dt$, where $Q \equiv A \sinh^2 t + B \sinh t + C$ and $V(t) \equiv \int (\nu \sinh t - m)/Q dt$:

$$J_{\text{arcsinh}} = \frac{1}{2} [t V(t)]_{t_0}^{t_1} - \frac{1}{2} \int_{t_0}^{t_1} V(t) dt. \quad (7.28)$$

The poles β_k . Substitute $y = e^t$ in the integrand of (7.27): with $\sinh t = \frac{1}{2}(y - 1/y)$ and $dt = dy/y$ one has $\nu \sinh t - m = N(y)/(2y)$ and $Q = D(y)/(4y^2)$, so

$$\frac{\nu \sinh t - m}{Q} dt = 2R(y) dy, \quad R(y) \equiv \frac{N(y)}{D(y)}, \quad (7.29)$$

$$N(y) = \nu(y^2 - 1) - 2my, \quad D(y) = A(y^4 + 1) + 2B(y^3 - y) + (4C - 2A)y^2,$$

and $V = \int 2R dy$. The four *poles* β_k of R are the roots of D : $Q = 0$ means $\sinh t = \zeta$ with ζ a root of $A\zeta^2 + B\zeta + C$, and since $B^2 - 4AC = -4w^2$ ($w \equiv \sqrt{1 + m^2 + \nu^2}$),

$$\zeta = \frac{-m\nu \pm iw}{1 + m^2}, \quad w = \sqrt{1 + m^2 + \nu^2}. \quad (7.30)$$

Each $\sinh t = \zeta$ is $y^2 - 2\zeta y - 1 = 0$, i.e. $y = \zeta \pm \sqrt{\zeta^2 + 1}$, so the four roots of D are these for the two ζ . For the upper root $\zeta^2 + 1 = (mw - i\nu)^2/(1 + m^2)^2$, hence $\sqrt{\zeta^2 + 1} = (mw - i\nu)/(1 + m^2)$ and $\beta = \zeta \pm \sqrt{\zeta^2 + 1}$ collapses with no complex square root to the two poles with $\text{Im } \beta > 0$,

$$\beta_+ = \frac{(w - \nu)(m + i)}{1 + m^2}, \quad \beta_- = \frac{(w + \nu)(-m + i)}{1 + m^2} \quad (7.31)$$

the remaining two being the conjugates $\bar{\beta}_\pm$ (with $\text{Im} < 0$). Write $a_\pm = \text{Re } \beta_\pm = \frac{\pm m(w \mp \nu)}{1 + m^2}$ and $b_\pm = \text{Im } \beta_\pm = \frac{w \mp \nu}{1 + m^2}$, all real.

The residues α_k are $\pm \frac{i}{4}$. $\alpha_k \equiv \text{Res}_{y=\beta_k} R = N(\beta_k)/D'(\beta_k)$ is the residue of R at the pole β_k . On the factored denominator $D = A(y^2 - 2\zeta_1 y - 1)(y^2 - 2\zeta_2 y - 1)$, a pole $\beta = \zeta_i \pm r_i$ ($r_i = \sqrt{\zeta_i^2 + 1}$) obeys $\beta^2 = 2\zeta_i \beta + 1$, so $N(\beta) = 2\beta(\nu\zeta_i - m)$ and $D'(\beta) = 4A(\pm r_i)(\zeta_i - \zeta_j)\beta$, giving

$$\alpha = \frac{N(\beta)}{D'(\beta)} = \frac{\nu\zeta_i - m}{2A(\pm r_i)(\zeta_i - \zeta_j)} = \pm \frac{i}{4},$$

purely imaginary, because $\nu\zeta_i - m = -w r_i$ (the $mw - i\nu$ of r_i cancels the numerator) and $\zeta_1 - \zeta_2 = 2iw/(1 + m^2)$. Since R is real, its residues come in conjugate pairs; tracking the branch in the residue above, the upper poles carry $\alpha(\beta_+) = +\frac{i}{4}$ and $\alpha(\beta_-) = -\frac{i}{4}$, and the lower ones the conjugates $\alpha(\bar{\beta}_\pm) = \mp \frac{i}{4}$. So $V = \int 2R dy = 2 \sum_{k=1}^4 \alpha_k \ln(y - \beta_k)$, and grouping each pole with its conjugate, the β_+ pair is

$$2 \left[\frac{i}{4} \ln(y - \beta_+) - \frac{i}{4} \ln(y - \bar{\beta}_+) \right] = \frac{i}{2} \ln \frac{y - \beta_+}{y - \bar{\beta}_+}.$$

For real y , $y - \bar{\beta}_+ = \overline{y - \beta_+}$, so the ratio equals $(y - \beta_+)/\overline{(y - \beta_+)} = e^{2i \arg(y - \beta_+)}$ (unit modulus) and its log is $2i \arg(y - \beta_+)$. The $\ln |\cdot|$ parts cancel and the pair collapses to $\frac{i}{2} \cdot 2i \arg(y - \beta_+) = -\arg(y - \beta_+)$. The β_- pair (residue $-\frac{i}{4}$) gives $+\arg(y - \beta_-)$ the same way, so, with $y - \beta_\pm = (y - a_\pm) - ib_\pm$ and hence $\arg(y - \beta_\pm) = \text{atan2}(-b_\pm, y - a_\pm)$,

$$V(t) = \arg(y - \beta_-) - \arg(y - \beta_+) = \text{atan2}(-b_-, y - a_-) - \text{atan2}(-b_+, y - a_+), \quad y = e^t. \quad (7.32)$$

The remaining integral. $\arg(y - \beta) = \text{Im} \ln(y - \beta)$, and with $y = e^t$ ($dt = dy/y$), splitting $\ln(y - \beta) = \ln(-\beta) + \ln(1 - y/\beta)$ gives $\int \ln(y - \beta) dy/y = \ln(-\beta) \ln y - \text{Li}_2(y/\beta)$, so

$$\begin{aligned} \int \arg(y - \beta) dt &= \ln y \arg(-\beta) - \text{Im} \text{Li}_2(y/\beta), \\ \int V dt &= \ln y [\arg(-\beta_-) - \arg(-\beta_+)] - [\text{Im} \text{Li}_2(y/\beta_-) - \text{Im} \text{Li}_2(y/\beta_+)]. \end{aligned} \quad (7.33)$$

Assembly. In (7.28), $t = \ln y$ so $tV = \ln y [\arg(y - \beta_-) - \arg(y - \beta_+)]$; combining $\frac{1}{2}[tV] - \frac{1}{2} \int V dt$ and using $\arg(y - \beta) - \arg(-\beta) = \arg(1 - y/\beta)$ on the two $\ln y$ pieces gives the manifestly real

$$J_{\text{arcsinh}} = -\frac{1}{2} [\text{Im} G(\beta_+) - \text{Im} G(\beta_-)]_{Y_0}^{Y_1}, \quad G(\beta) \equiv \text{Li}_2\left(\frac{y}{\beta}\right) + \ln y \ln\left(1 - \frac{y}{\beta}\right), \quad (7.34)$$

with $\text{Im} G(\beta) = \text{Im} \text{Li}_2(y/\beta) + \ln y \arg(1 - y/\beta)$, $Y_0 = e^{\text{arcsinh}(X_0)}$, $Y_1 = e^{\text{arcsinh}(X_1)}$.

Making it real: Bloch–Wigner and Clausen. The one transcendental left is $\text{Im} \text{Li}_2$. The Bloch–Wigner function $D(z) \equiv \text{Im} \text{Li}_2(z) + \arg(1 - z) \ln |z|$ is real and single-valued; with $z = y/\beta$ and $\ln |z| = \ln y - \ln |\beta|$ the $\ln y$ cancels, leaving

$$\text{Im} G(\beta) = D\left(\frac{y}{\beta}\right) + \ln |\beta| \arg\left(1 - \frac{y}{\beta}\right). \quad (7.35)$$

Everything is now elementary plus D . From (7.31), $\arg \beta_+ = \psi \equiv \text{atan2}(1, m)$, $\arg \beta_- = \pi - \psi$, and $|\beta_+| = (w - \nu)/\sqrt{1 + m^2} = 1/|\beta_-|$, so

$$\begin{aligned} \psi &= \text{atan2}(1, m), \quad L \equiv \ln |\beta_+| = \ln(w - \nu) - \frac{1}{2} \ln(1 + m^2) = -\ln |\beta_-|, \\ \varphi_{\pm} &\equiv \arg\left(1 - \frac{y}{\beta_{\pm}}\right) = \text{atan2}(y, w \mp \nu \mp my). \end{aligned} \quad (7.36)$$

D reduces to the Clausen function Cl_2 by Lewin, $D(re^{i\theta}) = \frac{1}{2}[\text{Cl}_2(2\theta) + \text{Cl}_2(2\omega) - \text{Cl}_2(2\theta + 2\omega)]$, $\omega = -\arg(1 - z)$; with $\theta = \arg(y/\beta_{\pm})$ and Cl_2 odd and 2π -periodic,

$$\begin{aligned} D\left(\frac{y}{\beta_+}\right) &= \frac{1}{2} [\text{Cl}_2(2\psi + 2\varphi_+) - \text{Cl}_2(2\psi) - \text{Cl}_2(2\varphi_+)], \\ D\left(\frac{y}{\beta_-}\right) &= \frac{1}{2} [\text{Cl}_2(2\psi) - \text{Cl}_2(2\varphi_-) - \text{Cl}_2(2\psi - 2\varphi_-)]. \end{aligned} \quad (7.37)$$

Since $\text{Im} G(\beta_+) = D(y/\beta_+) + L\varphi_+$ and $\text{Im} G(\beta_-) = D(y/\beta_-) - L\varphi_-$, (7.34) becomes the all-real recipe

$$J_{\text{arcsinh}} = -\frac{1}{2} \left[D\left(\frac{y}{\beta_+}\right) - D\left(\frac{y}{\beta_-}\right) + L(\varphi_+ + \varphi_-) \right]_{Y_0}^{Y_1}, \quad (7.38)$$

the y -independent $\text{Cl}_2(2\psi)$ canceling across $Y_0 \rightarrow Y_1$ (eight Cl_2 calls total). Cl_2 is the only special function – reduce its argument to $[0, \pi]$ by oddness and 2π -periodicity, then sum $\theta - \theta \ln \theta + \sum_k c_k \theta^{2k+1}$ (energies.py_tCoreReal/_cl2; coefficients in notes/arcsinhClausen.nb). Verified end to end in notes/verify_realroute.py: V is the antiderivative of $(\nu \sinh t - m)/Q$ (finite difference $\sim 10^{-10}$), and (7.28)–(7.34) match the direct integral (7.27) to $\sim 10^{-16}$; the $\beta_{\pm}$, the residues $\pm \frac{i}{4}$, and the Cl_2 recipe are checked in notes/partial_fractions_arcsinh.py and notes/arcsinhClausen.nb.

No further collapse. The two poles satisfy $\beta_+ \beta_- = -1$, i.e. $\beta_- = -1/\beta_+$ (indeed $(m+i)(-m+i) = -(1+m^2)$ and $(w-\nu)(w+\nu) = 1+m^2$). This tempts one to fold $\text{Im} G(\beta_+) - \text{Im} G(\beta_-)$ into a single term through the Bloch–Wigner inversion $D(1/z) = -D(z)$. It fails: the inversion would need $y/\beta_- = 1/(y/\beta_+) = \beta_+/y$, whereas $y/\beta_- = -y\beta_+$ (equal only if $-y^2 = 1$). Numerically $D(y/\beta_-) \neq \pm D(y/\beta_+)$ (off by $O(1)$), the two dilog arguments y/β_+ and $-y\beta_+$ are not a $z, -z$ duplication pair either, and $\arg(1 - y/\beta_+) + \arg(1 - y/\beta_-)$ is not constant – so no elementary relation ties them. The two Bloch–Wigner terms sit at genuinely unrelated points and (7.34) is the end of the line: it does not shorten. Checked in notes/verify_realroute.py.

8 Mollified gradient

This derivation repeats the work of §4 but with the new kernel. We start from (7.4) in the Φ_σ form,

$$A_\cap^\sigma = - \oint_{\partial A} d\ell_x \oint_{\partial B} d\ell_y \Phi_\sigma(|\vec{y} - \vec{x}|) (\hat{n}_A(\vec{x}) \cdot \hat{n}_B(\vec{y})).$$

Next we note that the line element is

$$\hat{n}_A^\alpha d\ell_x = \varepsilon_{\alpha\beta} dx^\beta, \quad (8.1)$$

and

$$n_A^\alpha d\ell_x n_B^\alpha d\ell_y = \varepsilon_{\alpha\beta} dx^\beta \varepsilon_{\alpha\gamma} dy^\gamma = \delta_{\beta\gamma} dx^\beta dy^\gamma = dx^\gamma dy^\gamma,$$

so that

$$A_\cap^\sigma = - \oint_{\partial A} dx^\alpha \oint_{\partial B} dy^\alpha \Phi_\sigma(|\vec{y} - \vec{x}|). \quad (8.2)$$

This is quite different from the situation in §4, but we apply the same technique. Let $\vec{x} \mapsto \vec{x} + \delta\vec{x}$, so that

$$\delta A_\cap^\sigma = - \oint_{\partial B} dy^\alpha \oint_{\partial A} (\delta\Phi_\sigma(|\vec{y} - \vec{x}|) dx^\alpha + \Phi_\sigma \delta(dx^\alpha)).$$

We know $\delta\Phi_\sigma = \frac{\partial\Phi_\sigma}{\partial x^\beta} \delta x^\beta$, therefore

$$\delta A_\cap^\sigma = - \oint_{\partial B} dy^\alpha \oint_{\partial A} \left[\frac{\partial\Phi_\sigma}{\partial x^\beta} \delta x^\beta dx^\alpha + \Phi_\sigma d\delta x^\alpha \right]. \quad (8.3)$$

As before, we integrate the second term by parts (the boundary term $\Phi_\sigma \delta x^\alpha$ vanishes on the closed loop), $\oint_{\partial A} d\delta x^\alpha \Phi_\sigma = - \oint_{\partial A} \delta x^\alpha \frac{\partial\Phi_\sigma}{\partial x^\beta} dx^\beta$, so

$$\delta A_\cap^\sigma = - \oint_{\partial A} \oint_{\partial B} \left[\delta x^\beta dx^\alpha \frac{\partial\Phi_\sigma}{\partial x^\beta} dy^\alpha - \delta x^\alpha dx^\gamma \frac{\partial\Phi_\sigma}{\partial x^\gamma} dy^\alpha \right].$$

These are all dummy indices; in the second term relabel $\alpha \rightarrow \beta$ and $\gamma \rightarrow \alpha$, giving

$$\delta A_\cap^\sigma = - \oint_{\partial A} dx^\alpha \oint_{\partial B} \left[dy^\alpha \frac{\partial\Phi_\sigma}{\partial x^\beta} - dy^\beta \frac{\partial\Phi_\sigma}{\partial x^\alpha} \right] \delta x^\beta. \quad (8.4)$$

We use (8.1) to put the $d\ell_y$ back in: $\varepsilon_{\alpha\beta} dy^\beta = \hat{n}_B^\alpha(\vec{y}) d\ell_y$; multiplying by $\varepsilon_{\gamma\alpha}$ gives $\varepsilon_{\gamma\alpha} \varepsilon_{\alpha\beta} dy^\beta = -dy^\gamma = \varepsilon_{\alpha\gamma} \hat{n}_B^\alpha d\ell_y$. Substituting,

$$\delta A_\cap^\sigma = - \oint_{\partial A} dx^\alpha \oint_{\partial B} \left[\varepsilon_{\gamma\alpha} \hat{n}_B^\gamma d\ell_y \frac{\partial\Phi_\sigma}{\partial x^\beta} - \varepsilon_{\gamma\beta} \hat{n}_B^\gamma d\ell_y \frac{\partial\Phi_\sigma}{\partial x^\alpha} \right] \delta x^\beta,$$

and the bracket is antisymmetric in $\alpha\beta$, so

$$\delta A_\cap^\sigma = - \oint_{\partial A} dx^\alpha \oint_{\partial B} d\ell_y \varepsilon_{\alpha\beta} \hat{n}_B^\gamma \frac{\partial\Phi_\sigma}{\partial x^\gamma} \delta x^\beta. \quad (8.5)$$

From (6.9) the inner integral is $\Psi_B^\sigma(\vec{x})$, hence

$$\delta A_\cap^\sigma = - \oint_{\partial A} dx^\alpha \delta x^\beta \varepsilon_{\alpha\beta} \Psi_B^\sigma(\vec{x}). \quad (8.6)$$

We traverse the edges of this outer integral over ∂A . Parameterizing as before, $\vec{x}_k(s) = \vec{v}_k + s\vec{e}_k = (1-s)\vec{v}_k + s\vec{v}_{z(k)}$ and $dx_k^\alpha = e_k^\alpha ds$, so

$$\delta A_\cap^\sigma = - \sum_{k \in \partial A} \sum_\beta \int_0^1 e_k^\alpha ds [(1-s)\delta v_k^\beta + s\delta v_{z(k)}^\beta] \varepsilon_{\alpha\beta} \Psi_B^\sigma(\vec{x}_k(s)).$$

If we let

$$W_k^0 \equiv \int_0^1 ds (1-s) \Psi_B^\sigma(\vec{x}_k(s)), \quad (8.7)$$

$$W_k^1 \equiv \int_0^1 ds s \Psi_B^\sigma(\vec{x}_k(s)), \quad (8.8)$$

then $\delta A_\cap^\sigma = - \sum_{k \in \partial A} \sum_\beta [W_k^0 \delta_{km} + W_k^1 \delta_{z(k)m}] \varepsilon_{\alpha\beta} e_k^\alpha \delta v_m^\beta$, meaning

$$\frac{\partial A_\cap^\sigma}{\partial v_m^\beta} = \varepsilon_{\beta\alpha} [W_m^0 e_m^\alpha + W_{z'(m)}^1 e_{z'(m)}^\alpha]. \quad (8.9)$$

It remains to evaluate Ψ_B^σ . Recall (6.8), $\Psi_B^\sigma(\vec{x}) = \oint_{\partial B} \vec{E}_\sigma(\vec{y} - \vec{x}) \cdot \hat{n}_B(\vec{y}) d\ell_y$, and again use $\hat{n}_B^\alpha d\ell_y = \varepsilon_{\alpha\beta} dy^\beta$ to write $\Psi_B^\sigma(\vec{x}) = \oint_{\partial B} E_\sigma^\alpha(\vec{y} - \vec{x}) \varepsilon_{\alpha\beta} dy^\beta$. Parameterizing edge k of ∂B , with $\vec{w}_k = \vec{x} - \vec{v}_k$,

$$\vec{y}_k(s) - \vec{x} = \vec{v}_k + \vec{e}_k s - \vec{x} = -\vec{w}_k + \vec{e}_k s, \quad d\vec{y}_k = \vec{e}_k ds,$$

so, using $\vec{E}_\sigma$,

$$\Psi_k^\sigma(\vec{x}) = \int_0^1 ds \frac{-w_k^\alpha + e_k^\alpha s}{2\pi(|\vec{w}_k - \vec{e}_k s|^2 + \sigma^2)} \varepsilon_{\alpha\beta} e_k^\beta. \quad (8.10)$$

Doesn't (8.10) look familiar? Since $(\vec{w}_k - \vec{e}_k s)^2 = |\vec{w}_k|^2 - 2\vec{w}_k \cdot \vec{e}_k s + |\vec{e}_k|^2 s^2$, and like before

$$a_k \equiv |\vec{e}_k|^2, \quad b_k \equiv -2\vec{w}_k \cdot \vec{e}_k, \quad c_k \equiv |\vec{w}_k|^2 + \sigma^2,$$

the $\vec{e}_k s$ piece of the numerator drops ($\vec{e}_k \times \vec{e}_k = 0$), leaving $\Psi_k^\sigma(\vec{x}) = - \int_0^1 ds \frac{w_k^\alpha \varepsilon_{\alpha\beta} e_k^\beta}{2\pi(a_k s^2 + b_k s + c_k)}$.

With $D_k = \sqrt{4a_k c_k - b_k^2}$, the s -integral is a single arctangent, so

$$\Psi_k^\sigma(\vec{x}) \equiv - \frac{w_k^\alpha \varepsilon_{\alpha\beta} e_k^\beta}{\pi D_k} \left[\arctan \frac{2a_k + b_k}{D_k} - \arctan \frac{b_k}{D_k} \right], \quad (8.11)$$

and

$$\Psi_B^\sigma(\vec{x}) = \sum_{k \in \partial B} \Psi_k^\sigma(\vec{x}). \quad (8.12)$$

Transcribed from the handwritten §8. The gradient (8.9) equals the analogue of the sharp force (4.3) with weight Ψ_B^σ ; I checked it against a central finite difference of (8.2) on two overlapping unit squares ($\sigma = 0.15$) to $\sim 10^{-8}$ at all vertices, and (8.11) against the direct $\oint_{\partial B} \vec{E}_\sigma \cdot \hat{n}_B d\ell_y$ to $\sim 2 \times 10^{-8}$. As $\sigma \rightarrow 0$, $\Psi_B^\sigma \rightarrow 1$ inside B , $\frac{1}{2}$ on an edge, (interior angle)/ 2π at a vertex, and 0 outside – the softened membership of $\vec{x}$ in B .

Closing W^0, W^1 : the fully-real gradient (my derivation). The moments (8.7), (8.8) are an outer $\int_0^1 ds$ over the edge $m \in \partial A$ of Ψ_B^σ , weighted by $1-s$ and s . Unlike the energy panel (7.13) (whose inner ∂B integral was the ln-panel (7.9)), here the inner integral is already the arctangent Ψ_k^σ (8.11), so the outer integrand is $\arctan \times (\text{rational})$. Fix an inner edge $k \in \partial B$, $\hat{e}_k = \vec{e}_k/|\vec{e}_k|$,

and resolve $\vec{x}_m(s) = \vec{v}_m + s\vec{e}_m$ along and across $\hat{e}_k$, defining the pair slope μ and scaled intercept ν (the §7 m, ν for this edge pair):

$$P_0 = (\vec{v}_m - \vec{v}_k) \cdot \hat{e}_k, \quad P_1 = \vec{e}_m \cdot \hat{e}_k, \quad X_0 = (\vec{v}_m - \vec{v}_k) \times \hat{e}_k, \quad X_1 = \vec{e}_m \times \hat{e}_k, \quad \mu \equiv \frac{P_1}{X_1}, \quad \nu \equiv \frac{U_0 - \mu X_0}{\sigma}, \quad \xi \equiv X_0 + sX_1$$

with $U_0 = P_0$ at $\vec{v}_k$ and $U_0 = P_0 - |\vec{e}_k|$ at $\vec{v}_k + \vec{e}_k$. Because $D_k = 2|\vec{e}_k|\sqrt{\xi^2 + \sigma^2}$ and $\vec{w}_k \times \vec{e}_k = |\vec{e}_k|\xi$, the two arctangents of (8.11) collapse to $\Theta(\xi) \equiv \arctan \frac{\mu\xi + \sigma\nu}{\sqrt{\xi^2 + \sigma^2}}$ at the two intercepts, so $\Psi_k^\sigma(\vec{x}_m(s)) = -\frac{\xi}{2\pi\sqrt{\xi^2 + \sigma^2}}[\Theta]$. Changing $s \mapsto \xi$ ($ds = d\xi/X_1$, $1 - s = (X_0 + X_1 - \xi)/X_1$), the moments become

$$W_m^1 = -\frac{[M_1' - X_0 M_1]_{X_0}^{X_0+X_1}}{2\pi X_1^2}, \quad W_m^0 = -\frac{[M_1]_{X_0}^{X_0+X_1}}{2\pi X_1} - W_m^1, \quad M_1 = \int \frac{\xi \Theta}{\sqrt{\xi^2 + \sigma^2}} d\xi, \quad M_1' = \int \frac{\xi^2 \Theta}{\sqrt{\xi^2 + \sigma^2}} d\xi.$$

One master for energy and gradient. M_1, M_1' and the energy panel's $I_{\tan}$ (§7) are the single family – integrate $\arctan \times \text{weight}$ by parts,

$$\mathbb{M}[V] \equiv \int V'(\xi) \Theta(\xi) d\xi = V(\xi) \Theta(\xi) - \int V(\xi) \Theta'(\xi) d\xi, \quad \Theta' = \frac{\mu\sigma^2 - \sigma\nu\xi}{\sqrt{\xi^2 + \sigma^2} Q_2},$$

$Q_2 = (1 + \mu^2)\xi^2 + 2\mu\sigma\nu\xi + \sigma^2(1 + \nu^2)$, differing only in the weight V' :

$V' = \sqrt{\xi^2 + \sigma^2}$	$V = v_+ \equiv \frac{1}{2}(\xi\sqrt{\xi^2 + \sigma^2} + \sigma^2 \operatorname{arcsinh} \frac{\xi}{\sigma})$	energy $I_{\tan}$
$V' = \xi/\sqrt{\xi^2 + \sigma^2}$	$V = \sqrt{\xi^2 + \sigma^2}$	gradient M_1
$V' = \xi^2/\sqrt{\xi^2 + \sigma^2}$	$V = v_- \equiv \frac{1}{2}(\xi\sqrt{\xi^2 + \sigma^2} - \sigma^2 \operatorname{arcsinh} \frac{\xi}{\sigma})$	gradient M_1'

In $\int V\Theta' d\xi$ the $\frac{1}{2}\xi\sqrt{\xi^2 + \sigma^2}$ of $v_\pm$ (over $\sqrt{\xi^2 + \sigma^2}$) leaves the elementary $\frac{1}{2}\varrho$, $\varrho \equiv \int \frac{\xi(\mu\sigma^2 - \sigma\nu\xi)}{Q_2} d\xi$; the $\sqrt{\xi^2 + \sigma^2}$ of M_1 leaves the elementary $\varsigma \equiv \int \frac{\mu\sigma^2 - \sigma\nu\xi}{Q_2} d\xi$; and the $\pm \frac{\sigma^2}{2} \operatorname{arcsinh}(\xi/\sigma)$ of $v_\pm$ leaves $\pm \frac{\sigma^2}{2} T$, $T \equiv \int \operatorname{arcsinh} \frac{\xi}{\sigma} \frac{\mu\sigma^2 - \sigma\nu\xi}{\sqrt{\xi^2 + \sigma^2} Q_2} d\xi$, so

$$M_1 = \sqrt{\xi^2 + \sigma^2} \Theta - \varsigma, \quad M_1' = v_- \Theta - \frac{1}{2}\varrho + \frac{\sigma^2}{2} T, \quad I_{\tan} \text{ core} = v_+ \Theta - \frac{1}{2}\varrho - \frac{\sigma^2}{2} T.$$

The one transcendental T is the §7 $\operatorname{arcsinh}$ integral: with $X = \xi/\sigma$, $T = -\int \frac{\operatorname{arcsinh} X (\nu X - \mu)}{\sqrt{X^2 + 1} (AX^2 + BX + C)} dX = -2J_{\operatorname{arcsinh}} (A = 1 + \mu^2, B = 2\mu\nu, C = 1 + \nu^2)$, closed by the fully-real recipe (7.27)–(7.38) at $(m, \nu) = (\mu, \nu)$. So the whole gradient is fully real: $W^0 (M_1)$ is elementary, and the only transcendental in W^1 is the *same* $J_{\operatorname{arcsinh}}$ that closes the energy – entering + there (v_+), – here (v_-).

Verified in `notes/master_form.py` (the three $\mathbb{M}[V]$ reductions, energy/ $W^0/W^1 = +J/0/-J$ to $\sim 10^{-16}$) and `notes/verify_gradient_masters.py` (M_1, M_1' and $T = -2J_{\operatorname{arcsinh}}$ to $\sim 10^{-14}$); the assembled W^0, W^1 (`energies.py` `wClosedVec`) match a central finite difference of $A_\cap^\sigma$ to $\sim 3 \times 10^{-9}$. Both W^1 (via $1/X_1^2$) and the energy panel amplify near-parallel edge pairs, so both need the bridge below.

Near-parallel bridge (my color). $X_1 = \vec{e}_m \times \hat{e}_k = |\vec{e}_m| \sin \theta_{mk}$ vanishes as the two edges align, and the energy panel's I_c, I_d carry the same degeneracy through the σ/e_β^y Jacobian of (7.17). There $\mu = P_1/X_1 \rightarrow \infty$ and the closed forms lose all precision to cancellation – yet each is a *finite* integral. Writing $\xi = X_0 + uX_1$ and undoing the X_1 division, the panel arctangent term (its I_c, I_d) and the two moment cores are

$$\int_0^1 \sqrt{\xi^2 + \sigma^2} \Theta du, \quad \int_0^1 \frac{\xi \Theta}{\sqrt{\xi^2 + \sigma^2}} du, \quad \int_0^1 \frac{u \xi \Theta}{\sqrt{\xi^2 + \sigma^2}} du, \quad \Theta = \arctan \frac{U_0 + P_1 u}{\sqrt{\xi^2 + \sigma^2}},$$

manifestly regular as $X_1 \rightarrow 0$, with limit the parallel branch. The only feature is the arctangent node $u_\star = -U_0/P_1$ (a peak of width $\sim \sigma$); splitting there, a 2×24 Gauss rule holds $\sim 10^{-15}$ down to $\sigma \sim 10^{-3}$. Used for $|X_1| \leq 10^{-2}|\vec{e}_m|$, closed forms elsewhere (`energies.py` `_ceBridge`, `_wBridge`). Verified against a dense 2-D quadrature of $\iint \ln(|\vec{x} - \vec{y}|^2 + \sigma^2)$: panel and moments hold $\sim 10^{-16}$ across the near-parallel sweep (was 4×10^{-6} at $|X_1| = 10^{-6}$ pre-bridge). At the relaxed $N=6$ grazing config ($\min |\sin \theta_{mk}| \sim 9 \times 10^{-6}$) the exact energy self-consistency improves $5 \times 10^{-2} \rightarrow 7 \times 10^{-9}$ and the gradient $5 \times 10^{-3} \rightarrow 3 \times 10^{-9}$; Newton on the fully-analytic force then reaches $\max |F| \sim 9 \times 10^{-13}$ in 4 steps, where it stalled at $\sim 10^{-2}$ before.